



RACEDAY API

Endpoint Plan — Student Edition

A clean, role-based REST API for managing races, enrolments, and results.

Two Roles. One Platform.

ORGANISER	Creates events, manages categories, captures results, and views all enrolments.
PARTICIPANT	Browses events, enters races, and tracks personal enrolments and results.

HTTP method colour legend

GET read	POST create	PATCH update	DELETE remove
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1. Auth & Users

Everything starts here. Register, log in, and manage profiles. JWT tokens keep sessions secure.

Method	Endpoint	Who Can Use	What It Does
POST	/auth/register	Public	Create an account; defaults to Participant
POST	/auth/login	Public	Return JWT access and refresh tokens
POST	/auth/refresh	Any	Refresh an expired access token
POST	/auth/logout	Any	Invalidate the current session
GET	/users/me	Any	Get your own profile
PATCH	/users/me	Any	Update your own profile
GET	/users/:id	Organiser	View any user's profile
PATCH	/users/:id/role	Organiser	Promote or demote a user's role

2. Events

The heart of RaceDay. Organisers create and manage events; everyone can browse upcoming races.



Method	Endpoint	Who Can Use	What It Does
GET	/events	Any	List events; filter by type, date, or location
GET	/events/:id	Any	Get full details for one event
POST	/events	Organiser	Create a new event
PATCH	/events/:id	Organiser	Update event details
DELETE	/events/:id	Organiser	Cancel or delete an event
GET	/events/upcoming	Any	List upcoming events sorted by date
GET	/events/:id/enrolments	Organiser	View participants enrolled in an event
POST	/events/:id/publish	Organiser	Open enrolment for an event
POST	/events/:id/close	Organiser	Close enrolment for an event

Event fields

- name, description, date, location, distance, event_type (run, walk, or cycle)

3. Categories

Organisers define event categories. Everyone can browse the available options.

Method	Endpoint	Who Can Use	What It Does
GET	/categories	Any	List categories; filter by event_id
GET	/categories/:id	Any	Get details for one category
GET	/events/:id/categories	Any	List all categories for an event
POST	/categories	Organiser	Create a category for an event
PATCH	/categories/:id	Organiser	Update category details
DELETE	/categories/:id	Organiser	Remove a category

Category fields

- name (for example: Under 20, Senior, 10km, or 21km)
- age_group (optional)
- distance (optional)



- event_id (foreign key)

4. Event Enrolments

Participants secure race entries. Validation prevents duplicate or invalid enrolments.

Method	Endpoint	Who Can Use	What It Does
POST	/enrolments	Participant	Enter an event by selecting a category
GET	/enrolments	Participant	List your own event enrolments
GET	/enrolments/:id	Participant	View one of your enrolments
DELETE	/enrolments/:id	Participant	Cancel your own enrolment
GET	/events/:id/enrolments	Organiser	View all enrolments for an event
GET	/categories/:id/enrolments	Organiser	View enrolments in a category

Enrolment fields

- participant_id, event_id, category_id
- enrolment_date
- status: pending, confirmed, cancelled, or attended

5. Results

Organisers capture finish times and positions after events. Participants view their personal results.

Method	Endpoint	Who Can Use	What It Does
GET	/results	Participant	View your own results
GET	/results/:id	Participant	View one of your results
POST	/results	Organiser	Capture finish time and position
PATCH	/results/:id	Organiser	Update a participant's result
DELETE	/results/:id	Organiser	Remove a result entry
GET	/events/:id/results	Any	View all results for an event
GET	/categories/:id/results	Any	View results for a category



Method	Endpoint	Who Can Use	What It Does
GET	/events/:id/results/leaderboard	Any	View leaderboard sorted by position
POST	/events/:id/results/bulk	Organiser	Bulk upload participant results

Result fields

- participant_id, event_id, category_id
- finish_time in HH:MM:SS format and position
- status: finished, DNF, or DNS

6. Combined Views

Convenient summary endpoints bring event and participant data together.

Method	Endpoint	Who Can Use	What It Does
GET	/events/:id/participants	Organiser	View enrolments and results for an event
GET	/participants/:id/events	Participant	View events a participant has joined
GET	/participants/:id/results	Participant	View all results for a participant
GET	/dashboard/organiser	Organiser	Overview of events, enrolments, and results
GET	/dashboard/participant	Participant	Overview of upcoming events and results

Design Notes for Student Developers

1. Role Enforcement

Middleware inspects the JWT claim (role: organiser | participant). Routes under /me/ always resolve to the authenticated user; never trust a client-supplied ID for ownership.

2. Ownership Checks

- Participants can only cancel or view their own enrolments and results.
- Organisers have full access to events they created.
- Enforce ownership at query level: WHERE user_id = :authUserId.



3. Event Status Machine

LIFECYCLE

draft → published → in_progress → completed → cancelled

- Participants cannot enrol in cancelled or completed events.
- Organisers cannot modify closed events.

4. Category Validation

- Confirm that the event exists and is not completed.
- Require a unique category name within each event.
- Validate age and distance values.

5. Enrolment Workflow

- Participant selects an event and category.
- System confirms the event is open.
- Create an enrolment with pending or confirmed status.
- Prevent duplicate enrolments for the same user, event, and category.

6. Result Workflow

- Capture results only after the event.
- Validate finish time as HH:MM:SS.
- Allow automatic or manual position calculation.
- Support DNF (Did Not Finish) and DNS (Did Not Start).

7. Data Validation

- Use ISO date format: YYYY-MM-DDTHH:MM:SS.
- Allowed event types: run, walk, cycle.
- Suggested distances: 5km, 10km, 21.1km, 42.2km.

8. Cascade Rules

- When deleting an event, remove its categories, enrolments, and results.
- Consider soft deletion instead of permanent deletion.



Pro Tip for the Project

START SMALL

Build Auth + Events + Categories first. Once that vertical slice works, layer on Enrolments and Results.